

Introduction

Biological tissue function depends strongly on spatial organization and composition. This is particularly true for vascular networks, whose branching geometry, vessel diameters, and connectivity together determine how functional the vascularization is. Evaluating Pazopanib, a tyrosine kinase inhibitor whose antiangiogenic mechanism is expected to manifest as measurable changes in vascular structure, depends as a result on accurate extraction and subsequent quantification. These structures have parameters that are path dependent: vessel density, diameter distributions, branching frequency and changes in tortuosity. These are abstract higher-order properties of the network, and cannot be reliably recovered from 2D sections, nor from simple point wise evaluation. This 3D nature, combined with the limitations of slice-based methods, motivates the use of high-resolution 3D imaging able to discern small vessels inside of the tumors. For the dataset used in this work, contrast-enhanced Micro-CT scans were acquired using an agent that increases the attenuation of small blood vessels, with the particularities laid out in Figure 1.

Existing tools for analyzing Micro-CT data are dominated by proprietary paid software, and the available open-source alternatives lack robust solutions for extracting small-scale vasculature with its challenging specificities: low contrast, non-vessel-like structures, diffusion gradients, and vessels only a few voxels across. As a result, the vasculature in the provided dataset has resisted prior analysis attempts.

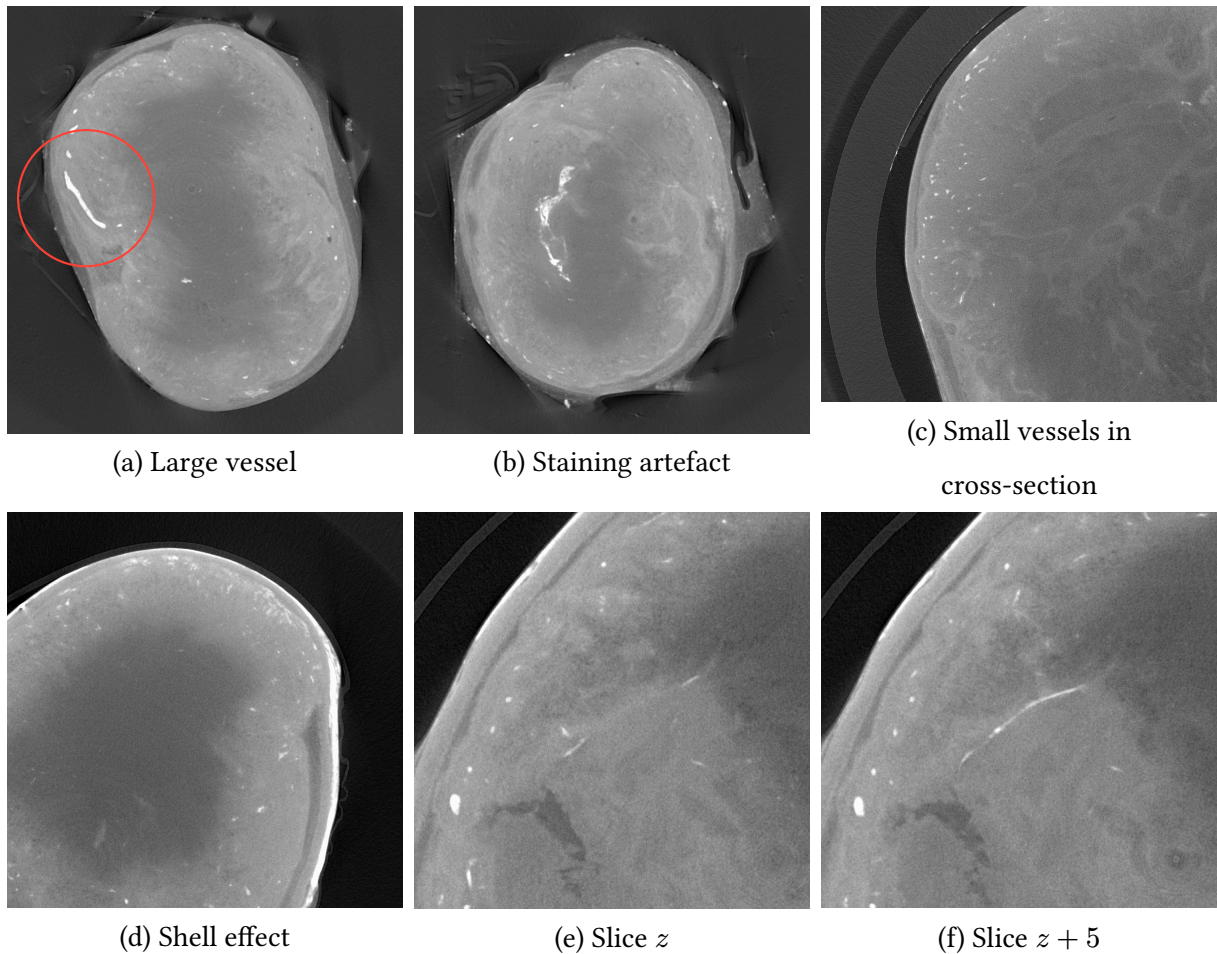


Figure 1: Slices illustrating the principal challenges: **(a)** A large high contrast vessel of approximately 14 voxels. **(b)** A large high-intensity non vessel-like structure resulting from hemorrhage. **(c)** Upper, left: many small vessels in cross-section, ranging from 2 to 8 voxels in diameter, where the partial-volume effect is present, as well as compression artifacts. **(d)** Outer surface: the “shell effect”, a high-intensity boundary surrounding the tumor caused by the diffusion of the contrast agent - also visible: a strong gradient between outside and center **(e, f)** Two slices from the same volume, separated by 5 voxels along the z -axis. The vessel indicated appears discontinuous in (e) but continuous in (f), highlighting the relevance of 3D methods.

Obtaining information from imaging

Analyzing tissue requires imaging it at a sufficient resolution to extract the structure of interest, and 3D imaging modalities leveraging tomography such as Micro-CT now make this possible at micrometer resolution. However 3D imaging carries with it challenges: a single high-resolution Micro-CT scan can reach into the thousands of slices, with large scans on the order of 3000^3 at 16 bits per pixel. This results in orders of magnitude more data than a comparable 2D acquisition, meaning that manual analysis as with 2D microscopy is impractical. Micro-CT data itself is also highly variable: scans differ on the same machine due to changes in imaged tissue, contrast agents and protocols used to make structures of interest visible, and further variation arises across machines and due to researchers' choices of acquisition parameters.

Analysis of this data is done using a fragmented landscape of closed source, paid proprietary software costing thousands, or open source software with variable quality, neither having fit for purpose microvasculature extraction tooling. Modern deep learning data driven techniques such as

machine learning require technical expertise and difficult to acquire annotated data, with poor out-of-distribution performance often demanding retraining.

Contribution and scope

To address the challenges of microvasculature extraction, an open-source pipeline called CollaboratiVessel is created to obtain segmented blood vessels, with the goal of enabling downstream extraction of clinically relevant vasculature characteristics. This goal is defined by the use case of a dataset of Contrast-Enhanced Micro-CT data acquired to examine the use of Pazopanib. The pipeline is created as an easily installable plugin for 3D Slicer, an open-source and free base software, that ingests and outputs data in portable data formats familiar to researchers, to ensure re-usability and compatibility with existing downstream software. The plugin implements proven algorithms with hyperparameter tuning handled by providing robust and tested standard parameters while keeping those relevant for common domain shifts in resolution and vessel size exposed, making the pipeline adaptable to new datasets. User feedback is leveraged in the loop by utilizing manually placed points and ease of use is prioritized to bridge the divide between computer science and researchers. Final performance is measured against a manually annotated baseline and thresholding, the previously used technique that did not succeed in segmenting with sufficient consistency for analysis.

All code written in the implementation of this thesis is made open-source under the MIT license, and the source of this document is labeled under the creative commons as CC0 - No Rights Reserved.